## Lab 2 Specifications

The lab 2 specifications are trying to get you to focus on two new topics: constructing large Verilog systems by intelligently reusing modules, and clean schematic construction.

### Hardware Specifications

#### Proficiency

- ☐ Resistors are in series with FPGA pins so that they correctly limit current
- ☐ Scanning signal appears on LEDs when appropriate buttons are depressed on the keypad.

#### Excellence

- ☐ Seven segment displays are same brightness regardless of how many segments are lit
- ☐ No noticeable bleeding of the digits between displays
- ☐ No flickering on the individual digits
- ☐ Scanning signal continues to be displayed under different keypress conditions, including multiple touches on rows or columns (at minimum 2 simultaneous presses).

### Verilog Structure and Style Specifications

#### Proficiency

- ☐ HDL design includes only a single seven-segment decoder module that is time multiplexed.
- ☐ Top design only includes instances of HSOSC, seven-segment module, multiplexing counter module, scanning module and assign statements to implement multiplexing and scanning passthrough.
- ☐ Scanning module only includes instances of a counter and assign statements that convert from the counter output to output values.
- ☐ Scanning module is resettable and enable-able.
- ☐ No counters implemented inline, all of them are contained in modules.

#### Excellence

- ☐ All counter modules (scanning and multiplexing) are parametrized instances of the counter module from lab 1.

#### Simulation Specifications

#### Proficiency

- ☐ No repeat simulations from lab 1. Those modules have already been verified!
- ☐ QuestaSim simulations carried out (possibly manually) for the scanning module and the top module.
- ☐ All simulations captured as legible waveform screenshots in report.

#### Excellence

- ☐ Automatic testbench for the scanning module that demonstrates all four output transitions, enable and reset features.
- ☐ Automatic testbench for the top module that exercises multiplexing functionality and LED driving functionality.

#### Block Diagram Specs

#### Proficiency and Excellence

- ☐ Block diagram exists and a good faith effort was made to follow the specs below. Block diagram specs in full effect in lab 3.

#### Other Documentation

#### Proficiency

- ☐ Current draw/sink on all FPGA pins are below the currents specified in the recommended operating conditions. Claims are backed up by calculations and reference to the appropriate items on the datasheet.

#### Excellence

- ☐ No errors in the calculations (correct forward voltage read off the datasheet, proper voltages used, etc.)

### Schematic Specifications

#### Proficiency

- ☐ All pin names labeled
- ☐ All pin numbers labeled
- ☐ Crossing wires clearly identified as junction or unconnected
- ☐ Neat layout (e.g., clear organization and spacing)
- ☐ All parts labeled with part number
- ☐ All component values present

#### Excellence

- ☐ Standard symbols used for all components where applicable
- ☐ Signals "flow" from left to right where possible (e.g., inputs on left hand side, outputs on right hand side)
- ☐ Title block with author name, title, and date

### Lab Writeup

#### Proficiency and Excellence

- ☐ Documentation embedded in a well-structured, concise report with one section per major deliverable. Sections should match structure of other specs.
- ☐ Spoiler tags are used to hide screenshots and code. Some unspoilered entries (~1 waveform, essential code snippets) allowed if they are essential to the narrative, kept to a minimum.
- ☐ Writeup contains minimal spelling or grammar issues and any errors do not significantly detract from clarity of the writeup.
- ☐ Statement of whether the design meets all the requirements. If not, list the shortcomings.
- ☐ AI prototype attempted and some reflection is recorded.
- ☐ Number of hours spent working on the lab are included.
- ☐ (Optional) List comments or suggestions on what was particularly good about the assignment or what you think needs to change in future versions.

## FOR REFERENCE, NOT YET IN EFFECT

### Block Diagram Specifications

#### Proficiency

- ☐ Each module is represented by a block diagram
- ☐ Bare logic in all module is represented by appropriately drawn gates.

- ☐ Submodules in each module are represented by blocks inside of the instantiating module.
- ☐ Submodules have all ports represented by pin symbols on the boundary of the submodule, each with a port name.
- ☐ Submodules indicate the name of the module being instantiated and the name of the instance.
- ☐ All connections between module ports, logic and submodules are indicated.
- ☐ Module ports are represented by pin symbols on the boundary of the module and labeled with port names
- ☐ All nets internal to the module are labeled, even when they share a name with a port on a module/submodule boundary.

### Excellence

- ☐ Appropriate use of bus notation for multi-bit signals (e.g., `data[7:0]` for an 8-bit bus), including fan-in and fan-out when needed.
- ☐ Appropriate use of implicit connections (i.e.: connect-by-label-name) to reduce routing connection
- ☐ Block diagram is an exact, 1:1 representation of the Verilog code.